Harshana Lakshara Fernando

Software Engineer | DevOps & Cloud Engineering

Panadura, Sri Lanka | +94 783 310 520 | harshanalf.work@gmail.com | Portfolio | LinkedIn | GitHub | Medium

Summary

Software Engineer with hands-on experience in Kubernetes, Docker, AWS, CI/CD, Kafka, Linux-based deployments, monitoring, and production support. Experienced in deploying distributed systems, automating delivery pipelines, troubleshooting production environments, and improving system reliability. CKA-trained, with strong Java, Spring Boot, and cloud-native engineering experience.

Skills

Cloud & DevOps	AWS | Kubernetes | Docker | Docker Compose | CI/CD | Jenkins | GitHub Actions | NGINX | Traefik
Infrastructure & Operations	Linux | Bash/Shell Scripting | Networking | Reverse Proxy | SSL/TLS | Production Support
Monitoring & Testing	Prometheus | Grafana | JMeter | Gatling | Logging | Performance Testing
Messaging & Distributed Systems	Kafka | RabbitMQ | gRPC | WebSockets | Protobuf | Resilience4j
Backend	Java | Spring Boot | Spring Cloud | Go | Gin | Node.js | REST APIs
Databases	PostgreSQL | Oracle | MySQL | Redis | Couchbase | DynamoDB

Experience

Software Engineer — Trading Systems *Sep 2026 – Present*
CAL | Colombo, Sri Lanka

- Building CAL's in-house electronic trading platform: a direct gateway to the **Colombo Stock Exchange (CSE)** plus the dealing, operations and testing tools around it.
- Led CSE's **188**-case **exchange certification**, building a Certification Runner that drives each case and explains every field difference against the expected result, backed by durable message logs and a CI-enforced test register.
- **Event-sourced CQRS** order path: a **Chronicle Queue** write-ahead journal with **PostgreSQL** as an async read model, so no database write sits between a trader and the exchange; the day's orders replay after a restart.
- **FIX 5.0 SP2** order routing and drop copy (**QuickFIX/J**), a custom **Java NIO** decoder for the **MITCH** UDP multicast feed (A/B feeds, snapshot recovery) and client-side rate limiting.
- Designed an **exchange simulator** (Python control plane, Java price-time matching engine) that replays **1,700+** real exchange conversations with **zero parity failures**, covering price bands, auctions and the trading-day schedule.
- Resilience and operations: FIX and market-data **failover** across primary and mirror hosts, sequence-number and feed-gap recovery, drop-copy **reconciliation** with an audited adopt/close path, **health-gated deployment** with rollback, and order-path changes behind default-off feature flags.
- Built the ops console (session control, reconciliation, chaos tests) and the **React** dealing screen, with pre-trade warnings for price bands, self-match and uncancellable AON orders.
- **~1,050 automated tests** (637 Java, 195 simulator, 216 console) in CI; every exchange behaviour traceable to the CSE spec or captured wire evidence.

R&D Software Engineer — KōreroCare Platform *Mar 2026 – Aug 2026*
Empathic Computing Laboratory, University of Auckland | Remote

- Containerized backend, database, authentication, and gateway services using **Docker Compose**.
- Configured **Traefik** routing for internal services and integrated **Keycloak OAuth2/OIDC** authentication.
- Managed environment configuration, service networking, health checks, and local deployment workflows.
- Designed **PostgreSQL**-backed backend services for high-volume mobile and wearable sensor ingestion.
- Supported secure service-to-service communication and deployment documentation for the research team.

Co-founder & Lead Engineer (Freelance) *Jul 2025 – Aug 2026*
ScaleX — SaaS Development Lab | Remote / Sri Lanka

- Delivered **AgriNova**, an inventory and commission platform on **Vercel** and **Supabase**/**PostgreSQL** across 5 branches and 329+ SKUs, tracking 194K+ units and 860+ invoices (Rs. 64.4M+).
- Reduced repeat dashboard data-fetch latency from ∼384ms to <1ms for cached views using **React Query** caching and pagination. Live demo: agrinova-demo.vercel.app.
- Owned end-to-end client-facing fixes, resolving production authentication, version compatibility, and workflow-level validation bugs.

Software Engineer Intern — Information Systems Associates (Pvt) Ltd *Dec 2024 – May 2026*
Aviation Technology | Colombo, Sri Lanka

- Deployed and supported a production Spring Boot and Kafka distributed service handling **25+ requests per second** with **<100ms latency**.
- Built Jenkins-based environment support and coordinated releases across development, QA, staging, and production environments.
- Created JMeter load tests and Prometheus/Grafana monitoring to track service latency, Kafka consumer behaviour, queue times, and order-processing performance.
- Diagnosed and resolved **Kafka consumer lag**, offset-reset, retry, and graceful-shutdown issues in production.
- Integrated **gRPC** and **Kafka** communication between a legacy Java 8 monolith and newer microservices.
- Supported phased production migration using feature toggles, rollback mechanisms, and a **strangler-pattern** deployment strategy.
- Maintained automated tests with **>90% unit test coverage** and collaborated with QA and distributed engineering teams.

Open Source Contributions — See all

- **KEDA #8317** (CNCF) *Merged*
  New **onlyReadyPods** option for the Kubernetes workload scaler (count only Running and Ready pods); design agreed with a maintainer; docs keda-docs #1919 merged Nov 2026.
- **kgateway #14851** (CNCF) *Merged, Oct 2026*
  Added the **kgateway_build_info** Prometheus metric (version, revision, Go version, OS/arch) so operators can see which build each pod runs; shipped in **kgateway v2.5.0-beta.2**.
- **kubernetes-mcp-server #1482** *Merged, Oct 2026*
  Fixed flaky HTTP server startup in the **Go** test suite with a 5 s readiness poll and cleanup before failing.
- **kubernetes-mcp-server #1549** *Merged, Oct 2026*
  Test that every tool, prompt and resource across all 12 toolsets declares valid **Kubernetes RBAC** metadata; approved on the first review.
- **Kubeflow Spark Operator #3249** *In review (LGTM)*
  20 **Ginkgo** specs covering every executor pod-phase transition: **100%** of `sparkpod_metrics.go`, plus a log message fix.
- **Jaeger #9997** (CNCF) *In review*
  Found and reported (#9991) a bug in Jaeger's **MCP** trace-topology tool: reused span IDs looped and parent cycles dropped spans; fix with 6 tests.
- **kgateway #14945** (CNCF) *In review*
  statPrefix for local rate limits on the traffic policy, with route template tokens; unit and translator tests.
- **Apache SkyWalking BanyanDB #1462** *In review*
  Found and fixed (#14199, assigned to me) case-sensitive **Bearer** auth in the **MCP** HTTP server; 4 tests; milestone 0.12.0.
- **Apache SkyWalking #14194** *Bug reported, Nov 2026*
  Found and reported **MCP** server config bugs (`.env` ignored, malformed values silently accepted) with a reproduction; fix by another contributor in banyandb #1460.
- **Apache Camel K #6933** *In review*
  Apache Pulsar autoscaler for Camel K's **KEDA** trait: scaler mapping, unit and e2e tests, docs.
- **kubernetes-mcp-server #1568** *Root-caused, Oct 2026*
  Showed a reported "flaky" test was a real merge-order bug; the author confirmed and applied the suggested fix.

Projects — See more projects on my portfolio

1. AgriNova Inventory & Commission Platform (2025 | Freelance, ScaleX) **Live demo**

Designed and delivered a production-grade inventory and sales management system for a real-world client, supporting multi-branch stock operations, invoicing, receipts, and commission tracking.

- Engineered transaction-safe stock movement logic and replaced application-layer aggregation with **PostgreSQL** aggregations, reducing execution time from **∼384 ms to ∼0.02 ms**.
- Designed a multi-layer caching strategy with **Next.js** and **React Query**, reducing repeated dashboard latency from **∼703 ms to ∼0.05 ms**.
- Built a clean, dynamic admin-focused UI and validated system reliability through end-to-end testing with **Playwright**.

Keywords: Next.js, PostgreSQL, Prisma, React Query, Tailwind CSS, Playwright

2. Employee Management System (EMS) (2025 | Individual) **Architecture | Github**

Designed and built a comprehensive, cloud-native microservices application for employee and department management.

- Containerized Spring Boot microservices and deployed them using **Kubernetes manifests** for deployments, services, configuration, and service discovery.
- Implemented CI/CD workflows using **GitHub Actions** and **ArgoCD**, with centralized configuration, API Gateway routing, Prometheus/Grafana monitoring, and Resilience4j fault tolerance.

Keywords: Spring Boot, Spring Cloud, Kubernetes, Docker, GitHub Actions, ArgoCD, Prometheus, Grafana, Resilience4j, API Gateway

3. Serverless Auction Platform (2025 | Individual) **Architecture | Github**

A capstone project built using AWS, following a microservices architecture with an event-driven, serverless (FaaS) design.

- Designed an AWS serverless architecture using **Lambda**, **API Gateway**, **DynamoDB**, **EventBridge**, **SQS**, and **SNS**, with asynchronous event processing and centralized error handling.
- Utilized **LocalStack** and **Docker** for local cloud emulation, ensuring reliable testing and deployment workflows.

Keywords: Node.js, Serverless Framework, AWS Lambda, DynamoDB, SQS, SNS, EventBridge, LocalStack, Docker

4. GoTogether (2024 | Group) **Architecture | Live | Github**

Developed and deployed a full-stack travel assistant platform with microservices in Next.js, Spring Boot, and Go.

- Deployed containerized Next.js, Spring Boot, Go, Keycloak, and PostgreSQL services behind **NGINX** on **Oracle Cloud**, with **Certbot**-managed HTTPS.
- Built a dedicated Auth Service using **Keycloak**, managing OAuth2/OIDC flows, token issuance, and automated realm/client boot-time import scripts.

Keywords: Next.js, Go, Spring Boot, Keycloak, PostgreSQL, NGINX, Docker, Oracle Cloud, Certbot

5. Event-Driven Food Delivery Platform (2024 | Group) **Github**

Built a scalable, event-driven food delivery platform using a modern microservices architecture and streaming.

- Enabled robust order synchronization using **Kafka**, processing asynchronous 'order_created' events across distributed microservices.
- Integrated **Redis caching** to improve API performance and reduce PostgreSQL database load, deploying the stack securely via **Docker**.

Keywords: Java, Spring Boot, Kafka, Redis, PostgreSQL, Docker, Microservices, Event-Driven

6. ZK-Rollup L2 Benchmarking Framework

Research Project — University of Moratuwa

- Built Docker-based benchmark orchestration scripts to evaluate Ethereum Layer-2 trade-offs across sequencing, batching, and data availability.
- Implemented the Rust-based Submitter service, observing **70% lower DA publication cost** using EIP-4844-style blobs versus calldata.
- Introduced mock-testable clients to improve testability and increased verifiable test coverage from **41% to 88%**.

Keywords: Rust, Docker, Python, L2 Rollups, Benchmarking

Education

B.Sc. (Hons) in Engineering – Computer Science and Engineering *2022 – 2026*
University of Moratuwa, Sri Lanka

G.C.E. Advanced Level (Physical Science Stream) *2019*
St. John's College, Panadura, Sri Lanka

Results: 3 A's (Combined Mathematics, Physics, Chemistry) **District Rank:** 17

Certifications & Professional Training

Certified Kubernetes Administrator preparation — KodeKloud, 2025 • AWS Solutions Architect Associate preparation — Udemy, 2025 • 100 Days of DevOps — KodeKloud, 2025 • Apache Kafka Series — Udemy, 2025 • Advanced OAuth Security — Udemy, 2024